

Angular HandsOn - 3

Directives & Pipes — Built-in and Custom

Aim:

To understand and implement Angular built-in structural directives (*ngIf, *ngFor, *ngSwitch), attribute directives (ngClass, ngStyle), custom directives, and pipes (built-in and custom) by developing a Student Course Portal application.

Codes:

src/app/pages/course-list/course-list.ts:

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { CourseCard } from '../../components/course-card/course-card';

@Component({
  selector: 'app-course-list',
  standalone: true,
  imports: [CommonModule, CourseCard],
  templateUrl: './course-list.html',
  styleUrls: ['./course-list.css']
})
export class CourseList {
  showCourses = true;
  department = 'IT';
  selectedCourseId = 0;
  courses = [
    {
      id: 1,
      name: 'Angular',
      code: 'ANG101',
      credits: 4,
      fee: 25000
    },
  ],
```

```

{
  id: 2,
  name: 'React',
  code: 'REA101',
  credits: 3,
  fee: 22000
},
{
  id: 3,
  name: 'NodeJS',
  code: 'NOD101',
  credits: 4,
  fee: 21000
}
];
onEnroll(id: number) {
  this.selectedCourseId = id;
}
}

```

src/app/pages/course-list/course-list.html:

```

<h2>Course List</h2>
<button (click)="showCourses = !showCourses">
  Toggle Courses
</button>
<br><br>
<div *ngIf="showCourses">
  <app-course-card
    *ngFor="let c of courses"

```

```

        [course]="c"
        (enrollRequested)="onEnroll($event)">
    </app-course-card>
</div>
<h3 *ngIf="selectedCourseId">
    Selected Course ID: {{ selectedCourseId }}
</h3>
<div [ngSwitch]="department">
    <p *ngSwitchCase="'IT'">
        Information Technology Department
    </p>
    <p *ngSwitchCase="'CSE'">
        Computer Science Department
    </p>
    <p *ngSwitchDefault>
        No Department Found
    </p>
</div>

```

src/app/components/course-card/course-card.ts:

```

import { Component, EventEmitter, Input, Output } from '@angular/core';
import { CommonModule } from '@angular/common';

@Component({
    selector: 'app-course-card',
    standalone: true,
    imports: [CommonModule],
    templateUrl: './course-card.html',
    styleUrls: ['./course-card.css']
})

```

```
export class CourseCard {

  @Input()
  course: any;

  @Output()
  enrollRequested = new EventEmitter<number>();

  enroll() {
    this.enrollRequested.emit(this.course.id);
  }
}
```

src/app/components/course-card/course-card.html:

```
<div
  [ngStyle]="{
    'background-color': '#f4f4f4',
    'padding': '15px',
    'margin-bottom': '10px',
    'border-radius': '8px',
    'border': '1px solid #ccc'
  }">
  <h3>{{ course.name | uppercase }}</h3>
  <p><strong>Code:</strong> {{ course.code }}</p>
  <p><strong>Credits:</strong> {{ course.credits }}</p>
  <p><strong>Fee:</strong> {{ course.fee | currency:'INR' }}</p>
  <button (click)="enroll()">
    Enroll
  </button>
</div>
```

src/app/components/course-card/course-card.css:

```
button {  
  background-color: #1976d2;  
  color: white;  
  border: none;  
  padding: 8px 15px;  
  cursor: pointer;  
  border-radius: 5px;  
}  
button:hover {  
  background-color: #1565c0;  
}  
h3 {  
  color: #1976d2;  
}
```

src/app/app.ts:

```
import { Component } from '@angular/core';  
import { CourseList } from '../pages/course-list/course-list';  
@Component({  
  selector: 'app-root',  
  standalone: true,  
  imports: [CourseList],  
  templateUrl: './app.html',  
  styleUrls: ['./app.css']  
})  
export class App {  
}
```

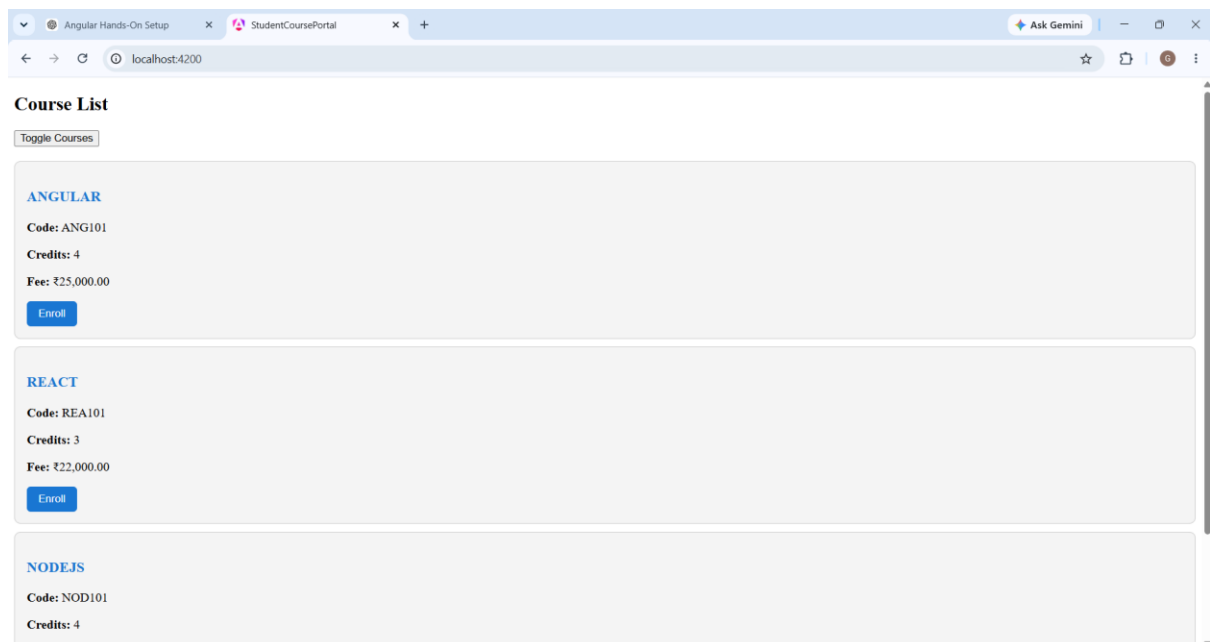
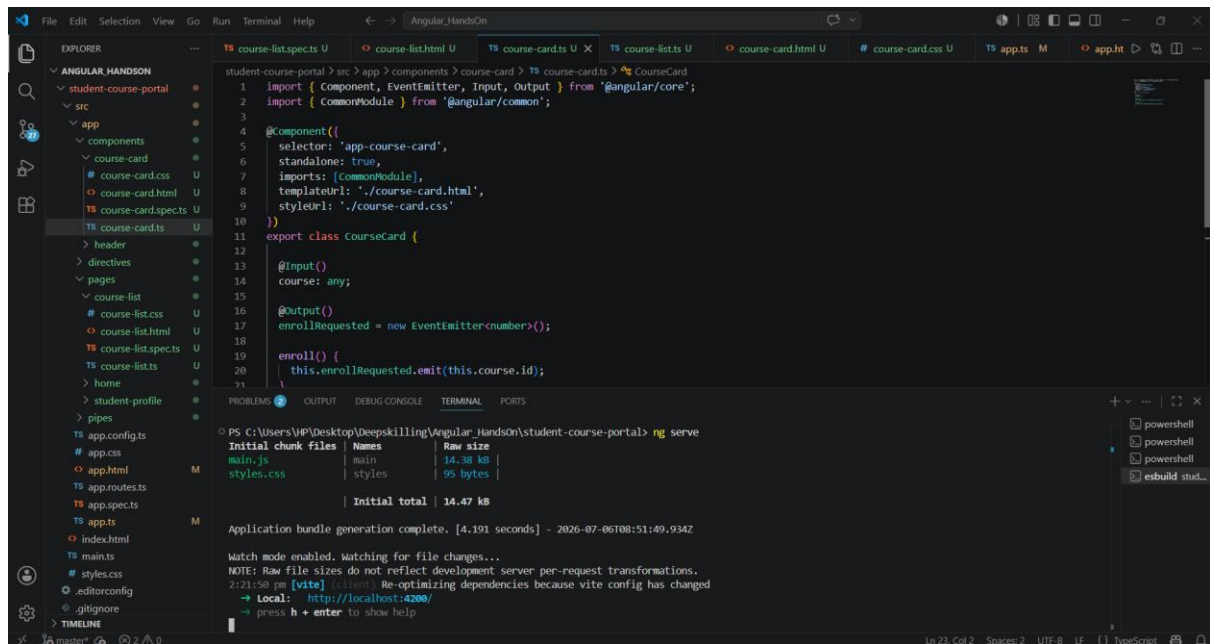
src/app/app.html:

<app-course-list></app-course-list>

Run the Project:

ng serve

Outputs:



Angular Hands-On SetupStudentCoursePortal

Ask Gemini

localhost:4200

Code: ANG101

Credits: 4

Fee: ₹25,000.00

Enroll

REACT

Code: REA101

Credits: 3

Fee: ₹22,000.00

Enroll

NODEJS

Code: NOD101

Credits: 4

Fee: ₹21,000.00

Enroll

Selected Course ID: 1

Information Technology Department

Angular Hands-On SetupStudentCoursePortal

Ask Gemini

localhost:4200

Course List

Toggle Courses

Selected Course ID: 3

Information Technology Department